MUHAMMAD SHAHBAZ

Full-Stack Software Engineer — Next.js · React · TypeScript · Node.js · AI Integrations

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PROFILE

Full-stack engineer who has built **three end-to-end applications** — an AI-powered gym management platform with 95+ API routes, a corporate web platform with a secure admin dashboard, and a cross-platform AI healthcare app — using Next.js, React, TypeScript, Node.js, Prisma, PostgreSQL, and MongoDB. Comfortable owning a project from schema design through deployment, with hands-on experience integrating OpenAI, Stripe, and OAuth into real products, plus an ML energy-demand pipeline with honest unseen-building evaluation. Also taught 100+ students as a programming instructor, translating complex engineering concepts into practical skills.

WORK EXPERIENCE

Programming Instructor | Aptech *Jun 2025 – Oct 2026*

- Delivered hands-on, project-based training in **JavaScript (ES6+), TypeScript, React.js, Node.js, Express.js, PHP, Laravel**, and modern CSS frameworks to 100+ beginner and intermediate students.
- Taught database design, SQL, and relational modeling; guided students through the full SDLC — planning, UI/UX, system design, and deployment — using the same workflows applied in production projects.
- Reviewed student code and mentored on clean architecture, debugging, testing, and Git workflows, strengthening the same code-quality practices applied in personal production projects.
- Designed capstone projects and assessments that simulated real-world development scenarios, and ran supplementary training in Excel (formulas, PivotTables, data analysis) for business applications.

PROJECTS

Titan Fitness — AI-Powered Gym Management Platform | Personal Project

Live Demo: titan-fitness-navy.vercel.app/

- Architected and built a production-scale platform spanning **16 member dashboard pages, 11 admin pages, and 95+ REST API routes** using Next.js, TypeScript, and Prisma/PostgreSQL.
- Implemented secure authentication (Better Auth, OTP verification, Google OAuth, role-based access control) and integrated **Stripe payments**, gamification, and push notifications.
- Built AI-powered coaching, workout generation, and nutrition planning via the OpenAI API, with rule-based fallback logic to guarantee availability when the AI service is unavailable.
- Shipped as a Progressive Web App (PWA) for installable, offline-capable mobile access.
- Stack: Next.js, React, TypeScript, Prisma, PostgreSQL, Better Auth, Stripe, OpenAI SDK, Tailwind CSS, React Query, Zustand.

BIOSAF Enterprises — Corporate Website & Admin Dashboard | Client Project

Live Demo: client-nine-gamma-23.vercel.app/

- Delivered a production website and secure admin dashboard for a commercial client, covering authentication, CRUD modules, and REST APIs for quote/contact management.
- Diagnosed and resolved routing and rendering performance issues, improving page load behavior and accessibility compliance across the site.
- Implemented SEO optimization and a fully responsive UI to support the client's public-facing marketing goals.
- Stack: Next.js, TypeScript, Prisma, MySQL, Tailwind CSS, Framer Motion, Zod.

SehatSathi AI — AI-Powered Healthcare Assistant | React Native / Expo

- Developed a cross-platform mobile health assistant with AI-powered chat, medical report analysis, medicine reminders, and emergency SOS — with full English/Urdu localization.
- Built secure authentication and cloud data sync using Firebase Auth and Firestore, plus family health profile management and push notifications.
- Stack: React Native, Expo, TypeScript, Firebase, Firestore, NativeWind, OpenAI API.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, PHP, Python, SQL, Rust, HTML5, CSS3
Frontend: React.js, Next.js, React Native (Expo), Redux, Zustand, TanStack Query, Tailwind CSS, Framer Motion
Backend: Node.js, Express.js, Hono, Laravel, REST APIs, JWT, Better Auth, MVC Architecture
Databases: PostgreSQL, MongoDB, MySQL, SQLite, Firebase, Prisma ORM, Drizzle ORM
AI, ML & Integrations: OpenAI API, AI SDK, scikit-learn, pandas, Streamlit, Stripe, WebSockets, Resend API
Testing & Tools: Zod, Jest, Vitest, Supertest, Git/GitHub, Docker, Vercel, Postman
Languages Spoken: English (Fluent), Urdu (Native), German (A2), Japanese (N5)

EDUCATION

Advanced Diploma in Software Engineering | Aptech *Nov 2022 – Jan 2025*

B.E. in Mechanical Engineering | NED University of Engineering and Technology *Oct 2023 – Jun 2027 (In Progress)*